

# Hamna Raeel

Software Engineer · ML/AI

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## Summary

Software Engineer with 3+ years of experience spanning ML/AI research, MLOps, and full-stack development. Builds and deploys deep learning systems – segmentation, classification, and RAG/LLM pipelines – and ships them through containerized, CI/CD-driven infrastructure across multi-cloud environments. Equally comfortable building the model and the product around it: production web applications with React, Next.js, FastAPI, and Node.js. 5 roles across research, DevOps, and engineering; 9+ projects shipped end-to-end.

## Technical Skills

**Languages:** Python, R, SQL, JavaScript, TypeScript, C++

**ML/DL Frameworks:** PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, Pandas, NumPy, Matplotlib

**Models & Architectures:** CNN/ResNet/UNet, Mask R-CNN, Mask2Former, ViT/Swin Transformer, DETR/SegFormer, mDeBERTa

**MLOps & Tools:** HuggingFace, Docker, Kubernetes, Jenkins CI/CD, Optuna, Weights & Biases, Git

**Cloud & Data:** IBM Cloud, Microsoft Azure, VMware, PostgreSQL, MySQL, Firebase, Prometheus, Grafana

**Full-Stack:** React, Next.js, Node.js, FastAPI, REST APIs, Tailwind CSS, Redux, Jest, React Native

## Work Experience

### Artificial Intelligence Diagnostic Lab, FAST-NUCES

Islamabad

*Research Assistant*

*Oct 2025 – Jan 2026*

- Trained deep learning segmentation models (Mask & Fast R-CNN, SegFormer, DETR) for histopathology analysis.
- Optimized preprocessing, augmentation, and training pipelines to boost model performance and efficiency.
- Built and evaluated classification models (ResNet, DenseNet, EfficientNet, Vision and Swin Transformer).
- Improved model performance through systematic feature extraction, hyperparameter optimization, and evaluation.

### Software Productivity Strategists, Inc. (SPS)

Islamabad

*DevOps and Cloud Infrastructure Intern*

*Jul 2025 – Sep 2025*

- Developed ELT data pipelines and automated CI/CD deployment workflows using Jenkins for reliable model releases.
- Containerized applications with Docker and orchestrated deployments using Kubernetes on IBM Cloud.
- Implemented monitoring and logging using Prometheus and Grafana for cloud-native applications.
- Managed deployments and infrastructure across multi-cloud environments, including Azure, IBM Cloud, and VMware.

### Artificial Intelligence Diagnostic Lab, FAST-NUCES

Islamabad

*Data Science Intern*

*Mar 2025 – May 2025*

- Implemented RAG pipelines using vector databases, embedding models, and LLMs for document Q&A.
- Developed and evaluated ML/DL models (CNN, ResNet, UNet, SVM, Decision Trees) across benchmark datasets.
- Applied ML to medical imaging and retail datasets, including CBIS-DDSM, Chest X-Ray, Retinal Vessel, and CIFAR-10/100.

### Optimus Bee

Lahore

*Software Engineer*

*Nov 2022 – Dec 2024*

- Developed a Mental Health Wellness application with an AI-powered chatbot, mood tracking, and recommendations.
- Built an e-commerce platform with integrated Stripe payment processing, authentication, and order management.
- Created a Financial Budget Management Tool for expense tracking, categorization, and analytics dashboards.
- Built an admin dashboard with dynamic UI, REST APIs, and reporting for an Australian travel agency.

### Cyber Legends

Lahore

*Front End Developer*

*Jun 2021 – Oct 2022*

- Led development of a WordPress-based university portal (NASTP-IIT) with a custom online admission system.
- Built "Infinite Writing," a freelance services platform with dual client/writer dashboards using HTML, Tailwind CSS, and JavaScript. [\[GitHub\]](#)

## Research & ML Projects

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### Segmentation-Guided Histopathology Analysis for Hirschsprung Disease Detection GitHub | Live Demo

- Built a Mask2Former-based segmentation model with a ResNet–Swin Transformer dual-backbone fusion, achieving a 92% Dice Score on ganglion cell segmentation.
- Designed a downstream classification framework on top of segmentation output, reaching 94% accuracy and 95.3% AUC-ROC for disease detection and deployed an inference demo on Hugging Face Spaces for interactive, real-time model evaluation.

### Meta-Learning for Zero-Shot Cross-Lingual Transfer (CML-mDeBERTa) GitHub | Live Demo

- Implemented a Curriculum Meta-Learning framework with first-order MAML on the XNLI dataset, enabling zero-shot natural language inference across 15 languages.
- Achieved 80.34% mean accuracy, an 11% improvement over baseline cross-lingual transfer methods and deployed an inference demo on Hugging Face Spaces

## Engineering & Full-Stack Projects

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### CloudETL – ELT pipelines with CI/CD, Docker, and Kubernetes on IBM Cloud GitHub

- Automated ELT data pipelines with Jenkins-driven CI/CD for fast, reliable, repeatable releases.
- Containerized services with Docker and orchestrated scalable deployments on Kubernetes clusters.
- Deployed, monitored, and managed containerized infrastructure and workloads on IBM Cloud.

### WebApp – Multi-cloud deployment with monitoring GitHub

- Deployed a production web application via CI/CD pipelines across Azure, IBM Cloud, and VMware.
- Implemented Prometheus and Grafana dashboards for real-time monitoring, logging, and alerts.
- Managed multi-cloud infrastructure to ensure consistent, reliable, zero-downtime deployments.

### Mental Health Wellness App – AI-powered chatbot and tracking GitHub

- Developed an AI-powered chatbot offering mental health support, guidance, and coping strategies.
- Built mood tracking features with personalized, data-driven wellness recommendations.

### Financial Budget Management Tool – Expense analytics GitHub

- Built expense tracking and categorization features for everyday personal finance management.
- Designed interactive analytics dashboards to visualize spending patterns and trends over time.
- Implemented budget goal-setting with real-time progress tracking and spending alerts.

### Travel Agency Admin Dashboard – Dynamic UI with REST APIs Live Site

- Built an admin dashboard with a dynamic, data-driven UI to streamline daily operations.
- Developed REST APIs for user management, booking workflows, and backend data operations.
- Implemented custom reporting features for an Australian travel agency’s admin team.

## Education

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### FAST, National University of Computer and Emerging Sciences (NUCES) Islamabad

*Master of Science in Data Science*

- Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI

## Certifications & Trainings

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- AI Fluency: Framework & Foundations – **Anthropic**
- Google Advanced Data Analytics – **Google**
- AI Concepts for Developers – **Microsoft**
- AI Apps & Agents on Azure – **Microsoft**
- ChatGPT Prompt-Engineering for Developers – **DeepLearning.AI**
- Databases and SQL for Data Science with Python – **IBM**